

| 5(a)                   | <table><tr><th>phenotype</th><th>observed</th><th>expected</th><th>O – E</th><th>(O – E)<sup>2</sup></th><th><math>\frac{(O - E)^2}{E}</math></th></tr><tr><td>red with black spots</td><td>279</td><td>281.25</td><td>–2.25</td><td>5.0625</td><td>0.018</td></tr><tr><td>white with black spots</td><td>95</td><td>93.75</td><td>1.25</td><td>1.5625</td><td>0.017</td></tr><tr><td>red</td><td>96</td><td>93.75</td><td>2.25</td><td>5.0625</td><td>0.054</td></tr><tr><td>white</td><td>30</td><td>31.25</td><td>–1.25 ;</td><td>1.5625 ;</td><td>0.05(0)</td></tr><tr><td colspan="5"></td><td><b>0.139 /<br/>0.14 ;</b></td></tr></table> <p>one mark for each column</p> | phenotype | observed | expected             | O – E                     | (O – E) <sup>2</sup> | $\frac{(O - E)^2}{E}$ | red with black spots | 279 | 281.25 | –2.25 | 5.0625 | 0.018 | white with black spots | 95 | 93.75 | 1.25 | 1.5625 | 0.017 | red | 96 | 93.75 | 2.25 | 5.0625 | 0.054 | white | 30 | 31.25 | –1.25 ; | 1.5625 ; | 0.05(0) |  |  |  |  |  | <b>0.139 /<br/>0.14 ;</b> | 3 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------------------|---------------------------|----------------------|-----------------------|----------------------|-----|--------|-------|--------|-------|------------------------|----|-------|------|--------|-------|-----|----|-------|------|--------|-------|-------|----|-------|---------|----------|---------|--|--|--|--|--|---------------------------|---|
| phenotype              | observed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | expected  | O – E    | (O – E) <sup>2</sup> | $\frac{(O - E)^2}{E}$     |                      |                       |                      |     |        |       |        |       |                        |    |       |      |        |       |     |    |       |      |        |       |       |    |       |         |          |         |  |  |  |  |  |                           |   |
| red with black spots   | 279                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 281.25    | –2.25    | 5.0625               | 0.018                     |                      |                       |                      |     |        |       |        |       |                        |    |       |      |        |       |     |    |       |      |        |       |       |    |       |         |          |         |  |  |  |  |  |                           |   |
| white with black spots | 95                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 93.75     | 1.25     | 1.5625               | 0.017                     |                      |                       |                      |     |        |       |        |       |                        |    |       |      |        |       |     |    |       |      |        |       |       |    |       |         |          |         |  |  |  |  |  |                           |   |
| red                    | 96                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 93.75     | 2.25     | 5.0625               | 0.054                     |                      |                       |                      |     |        |       |        |       |                        |    |       |      |        |       |     |    |       |      |        |       |       |    |       |         |          |         |  |  |  |  |  |                           |   |
| white                  | 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 31.25     | –1.25 ;  | 1.5625 ;             | 0.05(0)                   |                      |                       |                      |     |        |       |        |       |                        |    |       |      |        |       |     |    |       |      |        |       |       |    |       |         |          |         |  |  |  |  |  |                           |   |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |          |                      | <b>0.139 /<br/>0.14 ;</b> |                      |                       |                      |     |        |       |        |       |                        |    |       |      |        |       |     |    |       |      |        |       |       |    |       |         |          |         |  |  |  |  |  |                           |   |
| 5(b)                   | <p>any <b>two</b> from:</p> <p>accept null hypothesis (no mark)</p> <p>1 <math>\chi^2</math> value / 0.139 / 0.14, is lower than, the critical value / 7.815 ;</p> <p>2 the observed numbers are not significantly different to the expected numbers<br/>(at <math>p = 0.05</math>) ;</p> <p>3 any differences are due to chance ;</p> <p>allow ecf from 5(a)</p>                                                                                                                                                                                                                                                                                                               | 2         |          |                      |                           |                      |                       |                      |     |        |       |        |       |                        |    |       |      |        |       |     |    |       |      |        |       |       |    |       |         |          |         |  |  |  |  |  |                           |   |

| Question         | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks                            |                                  |                                  |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
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| 5(c)             | <table><tr><td></td><td colspan="4">female gametes</td></tr><tr><td>male gametes</td><td>X<sup>R</sup>B</td><td>X<sup>r</sup>B</td><td>X<sup>R</sup>b</td><td>X<sup>r</sup>b</td></tr><tr><td rowspan="3">X<sup>r</sup>B</td><td>X<sup>R</sup>X<sup>r</sup>BB</td><td>X<sup>r</sup>X<sup>r</sup>BB</td><td>X<sup>R</sup>X<sup>r</sup>Bb</td><td>X<sup>r</sup>X<sup>r</sup>Bb</td></tr><tr><td>female</td><td>female</td><td>female</td><td>female</td></tr><tr><td>red + black spots</td><td>white + black spots</td><td>red + black spots</td><td>white + black spots</td></tr><tr><td rowspan="3">Y<sup>R</sup>B</td><td>X<sup>R</sup>Y<sup>R</sup>BB</td><td>X<sup>r</sup>Y<sup>R</sup>BB</td><td>X<sup>R</sup>Y<sup>R</sup>Bb</td><td>X<sup>r</sup>Y<sup>R</sup>Bb</td></tr><tr><td>male</td><td>male</td><td>male</td><td>male</td></tr><tr><td>red + black spots</td><td>red + black spots</td><td>red + black spots</td><td>red + black spots</td></tr><tr><td rowspan="3">X<sup>r</sup>b</td><td>X<sup>R</sup>X<sup>r</sup>Bb</td><td>X<sup>r</sup>X<sup>r</sup>Bb</td><td>X<sup>R</sup>X<sup>r</sup>bb</td><td>X<sup>r</sup>X<sup>r</sup>bb</td></tr><tr><td>female</td><td>female</td><td>female</td><td>female</td></tr><tr><td>red + black spots</td><td>white + black spots</td><td>red + no spots</td><td>white + no spots</td></tr><tr><td rowspan="3">Y<sup>R</sup>b</td><td>X<sup>R</sup>Y<sup>R</sup>Bb</td><td>X<sup>r</sup>Y<sup>R</sup>Bb</td><td>X<sup>R</sup>Y<sup>R</sup>bb</td><td>X<sup>r</sup>Y<sup>R</sup>bb</td></tr><tr><td>male</td><td>male</td><td>male</td><td>male</td></tr><tr><td>red + black spots</td><td>red + black spots</td><td>red + no spots</td><td>red + no spots</td></tr></table> |                                  | female gametes                   |                                  |  |  | male gametes | X <sup>R</sup> B | X <sup>r</sup> B | X <sup>R</sup> b | X <sup>r</sup> b | X <sup>r</sup> B | X <sup>R</sup> X <sup>r</sup> BB | X <sup>r</sup> X <sup>r</sup> BB | X <sup>R</sup> X <sup>r</sup> Bb | X <sup>r</sup> X <sup>r</sup> Bb | female | female | female | female | red + black spots | white + black spots | red + black spots | white + black spots | Y <sup>R</sup> B | X <sup>R</sup> Y <sup>R</sup> BB | X <sup>r</sup> Y <sup>R</sup> BB | X <sup>R</sup> Y <sup>R</sup> Bb | X <sup>r</sup> Y <sup>R</sup> Bb | male | male | male | male | red + black spots | red + black spots | red + black spots | red + black spots | X <sup>r</sup> b | X <sup>R</sup> X <sup>r</sup> Bb | X <sup>r</sup> X <sup>r</sup> Bb | X <sup>R</sup> X <sup>r</sup> bb | X <sup>r</sup> X <sup>r</sup> bb | female | female | female | female | red + black spots | white + black spots | red + no spots | white + no spots | Y <sup>R</sup> b | X <sup>R</sup> Y <sup>R</sup> Bb | X <sup>r</sup> Y <sup>R</sup> Bb | X <sup>R</sup> Y <sup>R</sup> bb | X <sup>r</sup> Y <sup>R</sup> bb | male | male | male | male | red + black spots | red + black spots | red + no spots | red + no spots | 4 |
|                  | female gametes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  |                                  |                                  |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
| male gametes     | X <sup>R</sup> B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X <sup>r</sup> B                 | X <sup>R</sup> b                 | X <sup>r</sup> b                 |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
| X <sup>r</sup> B | X <sup>R</sup> X <sup>r</sup> BB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X <sup>r</sup> X <sup>r</sup> BB | X <sup>R</sup> X <sup>r</sup> Bb | X <sup>r</sup> X <sup>r</sup> Bb |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | female                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | female                           | female                           | female                           |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | red + black spots                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | white + black spots              | red + black spots                | white + black spots              |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
| Y <sup>R</sup> B | X <sup>R</sup> Y <sup>R</sup> BB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X <sup>r</sup> Y <sup>R</sup> BB | X <sup>R</sup> Y <sup>R</sup> Bb | X <sup>r</sup> Y <sup>R</sup> Bb |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | male                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | male                             | male                             | male                             |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | red + black spots                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | red + black spots                | red + black spots                | red + black spots                |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
| X <sup>r</sup> b | X <sup>R</sup> X <sup>r</sup> Bb                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X <sup>r</sup> X <sup>r</sup> Bb | X <sup>R</sup> X <sup>r</sup> bb | X <sup>r</sup> X <sup>r</sup> bb |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | female                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | female                           | female                           | female                           |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | red + black spots                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | white + black spots              | red + no spots                   | white + no spots                 |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
| Y <sup>R</sup> b | X <sup>R</sup> Y <sup>R</sup> Bb                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X <sup>r</sup> Y <sup>R</sup> Bb | X <sup>R</sup> Y <sup>R</sup> bb | X <sup>r</sup> Y <sup>R</sup> bb |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | male                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | male                             | male                             | male                             |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |
|                  | red + black spots                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | red + black spots                | red + no spots                   | red + no spots                   |  |  |              |                  |                  |                  |                  |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                   |                     |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                   |                   |                  |                                  |                                  |                                  |                                  |        |        |        |        |                   |                     |                |                  |                  |                                  |                                  |                                  |                                  |      |      |      |      |                   |                   |                |                |   |

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| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| 5(d)     | <p><i>mark as pairs</i></p> <p>1 allele <b>R</b> / dominant red allele, is on <b>Y</b> chromosome ;</p> <p>2 (so all) males <b>inherit</b>, dominant red allele / allele <b>R</b><br/> <b>or</b><br/> only <b>Y<sup>R</sup></b> is present in the gametes ;</p> <p>3 no, allele <b>r</b> / recessive white allele, on <b>Y</b> chromosome<br/> <b>or</b><br/> allele <b>r</b> only exists on the <b>X</b> chromosome ;</p> <p>4 (so) males never inherit, recessive white allele / allele <b>r</b> ;</p> | 2     |
| 5(e)     | <p>any <b>two</b> from:</p> <p>1 mutation ;</p> <p>2 detail of mutation ;</p> <p>3 crossing over ;</p> <p>4 (of) the <b>R</b> allele / dominant red allele, from a <b>Y</b> chromosome to an <b>X</b> chromosome ;</p>                                                                                                                                                                                                                                                                                   | 2     |

| Question | Answer | Marks |
|----------|--------|-------|
|----------|--------|-------|